

AUTOPILOT — Project Plan & Milestone Tracker

Bosch Future Mobility Challenge · December 2025 – March 2026

Status Legend

✓ Completed	Task fully implemented and validated
🔄 In Progress	Task actively under development
○ Planned	Scheduled for a future phase
⚠️ Delayed	Blocked by hardware/external factors

Phase 1 — Foundational Control Architecture

Period: December 2025 · Report date: December 22, 2025 · Focus: Hardware setup, core software architecture, basic perception

Task / Module	Start Date	End Date	Status	Notes
Architecture & Planning	Dec 1, 2025	Dec 22, 2025	✓ Completed	Software/hardware integration design
Object Detection & Tracking	Dec 10, 2025	Dec 22, 2025	✓ Completed	YOLOv8 deployment, iterative refinement ongoing
Stop Sign Detection & Response	Dec 10, 2025	Dec 22, 2025	✓ Completed	25 cm/s steady speed with a halt on detection
Lateral Control	Dec 22, 2025	Feb 6, 2026	○ Planned	Steering angle estimation, feedback loop
Path Planning Algorithms	Dec 22, 2025	Feb 6, 2026	○ Planned	A*, Dijkstra's, real-time dynamic adjustment
State Estimation & Localization	Dec 22, 2025	Feb 6, 2026	○ Planned	IMU-based position/orientation within track

Phase 2 — Real-Time Perception & Control Pipeline

Period: January – February 2026 · Report date: February 6, 2026 · Focus: Lane detection, PID motor control, traffic sign recognition

Task / Module	Start Date	End Date	Status	Notes
Straight Lane Detection	Jan 5, 2026	Feb 6, 2026	✓ Completed	Midpoint-based method, multiprocessing module with OpenCV
Curved Lane Detection	Jan 20, 2026	Mar 9, 2026	🔄 In Progress	Ongoing — lighting/shadow robustness challenges
Vehicle Control (PID Motor)	Jan 15, 2026	Feb 6, 2026	✓ Completed	PID-based motor control responding to lane perception
Real-Time Dashboard	Jan 15, 2026	Feb 6, 2026	✓ Completed	Visualization of perception and control states
Traffic Sign Detection Model	Jan 10, 2026	Feb 6, 2026	✓ Completed	All BFMC-required signs identified, dynamic behavior adaptation
ROS Architecture Integration	Jan 20, 2026	Feb 6, 2026	✓ Completed	Modular multiprocessing + ROS-compatible design

Phase 3 — Algorithm Optimization & Advanced Maneuvers

Period: February – March 2026 · Report date: March 9, 2026 · Focus: Curve/highway detection, intersection crossing, parallel parking

Task / Module	Start Date	End Date	Status	Notes
Curve Lane Detection (Advanced)	Feb 7, 2026	Mar 9, 2026	✓ Completed	ROI masking, Canny edges, polynomial fit, Kalman filter
Highway Lane Detection	Feb 7, 2026	Mar 9, 2026	✓ Completed	Deep learning, confidence scores, and multiple hypothesis tracking
Intersection Crossing	Feb 15, 2026	Mar 9, 2026	✓ Completed	Sensor fusion, finite state machine, priority queuing
Parallel Parking	Feb 15, 2026	Mar 9, 2026	✓ Completed	Reeds-Shepp curves, ultrasonic space detection, pure pursuit
Hardware Power Board Repair	Feb 20, 2026	TBD	⚠ Delayed	Power board failure — limited to simulation/virtual testing
On-Vehicle Integration Testing	Mar 9, 2026	TBD	🔄 In Progress	Pending hardware resolution for real-world validation

Master Milestone Summary — All Modules

Consolidated view of all deliverables across all three phases with deadlines and current status

Module	Phase	Deadline	Status
Architecture & Planning	Phase 1	Dec 22, 2025	✓ Completed
Object Detection (YOLOv8)	Phase 1	Dec 22, 2025	✓ Completed
Stop Sign Detection & Response	Phase 1	Dec 22, 2025	✓ Completed
Straight Lane Detection	Phase 2	Feb 6, 2026	✓ Completed
Vehicle Control (PID)	Phase 2	Feb 6, 2026	✓ Completed
Traffic Sign Detection	Phase 2	Feb 6, 2026	✓ Completed
Real-Time Dashboard	Phase 2	Feb 6, 2026	✓ Completed
ROS Architecture	Phase 2	Feb 6, 2026	✓ Completed
Curved Lane Detection	Phase 3	Mar 9, 2026	✓ Completed
Highway Lane Detection	Phase 3	Mar 9, 2026	✓ Completed
Intersection Crossing	Phase 3	Mar 9, 2026	✓ Completed
Parallel Parking	Phase 3	Mar 9, 2026	✓ Completed
Curved Lane Detection (Phase 2)	Phase 2→3	Mar 9, 2026	✓ Completed
On-Vehicle Integration Testing	Phase 3+	TBD	🔄 In Progress
Lateral Control	Ongoing	TBD	🔄 In Progress
Power Board Hardware Repair	Ongoing	TBD	⚠️ Delayed